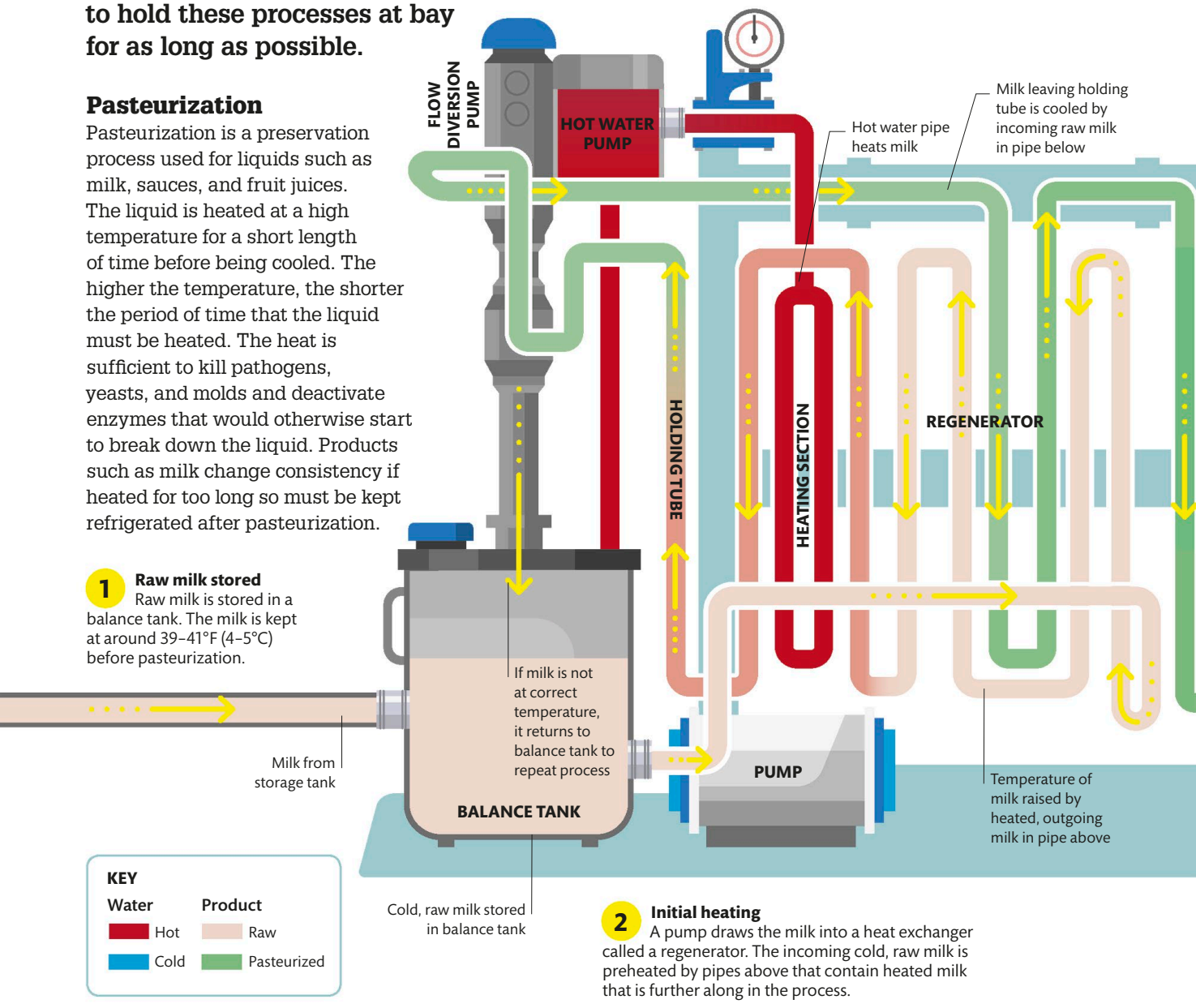
Food preservation

As soon as food is harvested, it is under attack from microscopic organisms, such as bacteria, and enzymes. These degrade the food until it becomes inedible. Over thousands of years, various methods have been developed to hold these processes at bay for as long as possible.

Pasteurization

Pasteurization is a preservation process used for liquids such as milk, sauces, and fruit juices. The liquid is heated at a high temperature for a short length of time before being cooled. The higher the temperature, the shorter the period of time that the liquid must be heated. The heat is sufficient to kill pathogens, yeasts, and molds and deactivate enzymes that would otherwise start to break down the liquid. Products such as milk change consistency if heated for too long so must be kept refrigerated after pasteurization.



1 Raw milk stored
Raw milk is stored in a balance tank. The milk is kept at around 39–41°F (4–5°C) before pasteurization.

2 Initial heating
A pump draws the milk into a heat exchanger called a regenerator. The incoming cold, raw milk is preheated by pipes above that contain heated milk that is further along in the process.

3 Secondary heating
The raw milk passes through a heating section, where pipes filled with hot water, supplied by a hot water pump, heat it further. The long, looped pipe ensures that the milk is kept at the correct temperature for long enough.

4 Heated milk checked
The milk flows into a holding tube, where it is kept for a period of time. A flow diversion pump at the top of the tube makes sure only pasteurized milk leaves. If the milk is hot enough, it begins the cooling process.